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Biomedical Engineering Refined Need Statements

Final Year Project Selection

September 18, 2026

Master List of Refined Need Statements

1. Women in active labour need a non-invasive method to measure cervical dilation to enable timely delivery decisions.
2. Healthcare workers in low-resource facilities need a rapid method to decontaminate reusable medical equipment to reduce cross-contamination.
3. Patients undergoing surgery need an affordable infection barrier method to reduce surgical-site infection rates.
4. Medical imaging technologists need personal radiation shielding to reduce cumulative occupational radiation exposure.
5. NICU healthcare workers need an efficient method to sterilize reusable medical tubing to reduce nosocomial infections.
6. Newborns with gastroschisis need rapid protection of exposed abdominal organs to prevent fatal infection.
7. High-risk surgical and critical care patients need an affordable method to maintain normothermia to prevent perioperative complications.
8. Radiographers in low-resource facilities need affordable digital imaging capture tools to lower radiography costs.
9. Children with mobility impairments need growth-accommodating assistive devices to maintain functional independence.
10. Children undergoing standing therapy need postural stabilization during weight-bearing exercises to prevent fall injuries.
11. Surgical patients need an affordable method to maintain body temperature to prevent perioperative hypothermia.
12. Neonates with hypoxic-ischemic encephalopathy need affordable therapeutic hypothermia to reduce brain injury.
13. Pregnant women at risk of preeclampsia need affordable continuous blood-pressure monitoring to detect hypertension early.
14. Patients experiencing significant hemorrhage need a rapid method to quantify blood loss to enable timely hemodynamic stabilization.
15. Patients at high risk of gastric cancer need non-invasive screening for early mucosal dysplasia to improve survival rates.
16. Pregnant women in underserved areas need low-cost screening to detect maternal anemia early.
17. Neonates requiring thermal care need continuous, low-resource temperature regulation to prevent thermal stress during care and transport.
18. Neonates receiving phototherapy need secure ocular shielding to prevent retinal damage.

19. Patients requiring urinary catheterization need secure catheter stabilization to prevent accidental displacement.
20. Patients with acute burns need extended-wear wound dressings to minimize dressing-change pain.
21. Elderly individuals with gait instability need continuous fall-prevention assist tools to prevent fall injuries.
22. Patients in cardiac arrest need continuous high-quality chest compressions to improve resuscitation survival.
23. Surgical trainees need affordable practical simulation tools to accelerate procedural skill acquisition.
24. Laboratory technicians performing genetic testing need rapid sequencing protocols to reduce turnaround time.
25. Manual wheelchair users traversing slopes need a method to prevent unintended rolling to reduce tipping and ejection accidents.
26. Patients with diabetic foot ulcers need early detection of tissue inflammation to prevent deep infection.
27. Newborns in regional hospitals need objective point-of-care hearing screening to identify congenital hearing loss early.
28. Neonates receiving CPAP need early detection of interface air leaks to maintain effective respiratory support.
29. Women experiencing prolonged labour need objective tracking of labour progression to detect dystocia early.
30. Neonates receiving enteral tube feeding need non-invasive verification of tube positioning to prevent pulmonary aspiration.
31. Children immobilized in casts need pressure monitoring beneath cast linings to prevent early tissue breakdown.
32. Children with hydrocephalus and shunts need non-invasive detection of shunt failure to enable prompt surgical referral.
33. Neonates receiving NG feeding need early detection of tube displacement to prevent pulmonary aspiration.
34. Patients recovering from major surgery need early detection of surgical wound dehiscence to enable immediate repair.
35. Patients undergoing prolonged surgery need continuous pressure distribution monitoring over vulnerable anatomy to prevent pressure injuries.
36. Patients with reduced lower-limb sensation need pressure monitoring beneath orthotic devices to prevent pressure ulcers.
37. Newborns at risk of congenital heart defects need point-of-care cardiac screening to enable early specialist referral.
38. Neonates in low-resource NICUs need rapid point-of-care screening for neonatal sepsis to enable timely antibiotic therapy.
39. Stroke survivors with upper-extremity paresis need targeted exercise tools to restore functional hand grasp.
40. Patients with acute burns need accurate assessment of burn depth to guide surgical debridement.
41. Trauma patients with suspected limb ischemia need rapid limb perfusion assessment to prevent tissue necrosis.
42. Cancer patients receiving chemotherapy need continuous nutritional tracking to detect malnutrition early.

43. Women undergoing IUD insertion need instant verification of correct device placement to prevent uterine perforation.
44. Newborns at risk of hypoglycemia need rapid blood glucose screening to prevent neurological damage.
45. Pregnant women living with HIV need continuous PMTCT tracking to prevent vertical HIV transmission.
46. Pediatric patients with physical disabilities need growth-accommodating mobility devices to maintain functional independence.
47. Children with clubfoot undergoing casting need growth-accommodating serial casts to maintain anatomical correction.
48. Trauma patients at risk of compartment syndrome need non-invasive compartment pressure monitoring to enable early fasciotomy.
49. Patients with suspected infection in low-resource settings need rapid differential testing between bacterial and viral pathogens to guide antibiotic stewardship.
50. Patients with suspected multi-pathogen infections need low-cost multiplex testing from a single sample to accelerate diagnosis.
51. Patients requiring urgent blood transfusion need rapid point-of-care blood compatibility verification to prevent hemolytic transfusion reactions.
52. Patients with suspected tuberculosis in rural areas need rapid point-of-care TB confirmation to initiate early therapy.
53. Cancer patients receiving chemotherapy need early infection detection to enable prompt treatment of neutropenic fever.
54. Cancer patients with chemotherapy-induced peripheral neuropathy need objective symptom monitoring to detect worsening nerve damage.
55. High-risk pregnant women in rural areas need continuous remote monitoring between antenatal visits to detect complications early.
56. Women living with obstetric fistula need accessible screening tools to enable early surgical referral.
57. Blood-bank staff need automated tracking of blood-product shelf life to minimize blood wastage.
58. Critically ill ICU patients need rapid point-of-care blood-gas analysis to guide immediate clinical decisions.
59. Premature and low-birth-weight infants need continuous thermal monitoring to detect hypothermia early.
60. Patients with chronic wounds in low-resource settings need low-cost wound management materials to accelerate healing.
61. Newborns in low-resource settings need affordable point-of-care sickle-cell screening to enable early diagnosis.
62. Elderly patients with hypertension need affordable home blood-pressure devices to facilitate regular monitoring.
63. Individuals with limb loss need affordable prosthetic devices to restore functional mobility.
64. Cancer patients in underserved settings need accessible psychosocial support tools to improve treatment adherence.
65. Patients receiving repeated chemotherapy need affordable long-term central venous access to reduce peripheral venipunctures.
66. Newborns in low-resource hospitals need low-cost precision weighing to guide nutritional management.
67. Patients in limited-lab settings need low-cost multiplex tests to detect multiple infections from one sample.

68. Healthcare workers in low-resource settings need affordable reusable PPE to maintain infection-control protection.
69. Children with hearing loss in low-resource settings need affordable, durable hearing aids to improve communication.
70. Individuals with upper-limb amputations need an affordable prosthetic control system to restore multi-articulating hand movement.
71. Neonates in transit need non-electric thermal maintenance to prevent hypothermia.
72. Newborns in hospitals without pediatric cardiologists need accessible CHD screening to enable early referral.
73. Pregnant women attending early antenatal care need accessible gestational age estimation to identify high-risk pregnancies early.
74. Children under five in drought-affected areas need rapid screening for severe acute malnutrition to enable timely nutritional rehabilitation.
75. Communities in cholera-prone areas need rapid testing for drinking water contamination to identify unsafe water sources.
76. Patients with diabetic foot ulcers need affordable offloading solutions to relieve wound pressure and promote healing.
77. People with diabetes need accessible retinal screening to detect diabetic retinopathy early.
78. Elderly patients in routine care need brief cognitive screening tools to detect early cognitive decline.
79. People affected by leprosy with reduced foot sensation need early detection of pressure injuries to prevent foot ulceration.
80. Stroke survivors living at home need accessible home rehabilitation tools to maintain long-term recovery.
81. People recently exposed to HIV need high-sensitivity early detection screening to enable prompt antiretroviral therapy.
82. Neonates with severe hyperbilirubinemia need non-invasive bilirubin measurement to reduce painful blood draws.
83. Patients with early malaria infection need low-density parasite detection to enable early treatment.
84. Patients undergoing major surgery need real-time blood loss measurement to guide fluid management.
85. Patients requiring venipuncture need affordable vein visualization to increase first-attempt success rates.
86. Biomedical engineers in low-resource hospitals need supply-chain access to local equipment replacement parts to reduce equipment downtime.
87. Laboratory staff in unstable grid regions need diagnostic equipment with power backup to prevent testing interruptions.
88. Clinical engineering teams need predictive maintenance monitoring for medical devices to eliminate unexpected equipment failure.
89. Patients receiving IV infusions need safe fluid warming prior to administration to maintain core temperature.
90. Infants with respiratory illness need rapid detection of acute respiratory distress to enable timely oxygen therapy.
91. Sterile processing staff need a scale-prevention method for steam sterilizers to maintain sterilizer uptime.
92. Surgical teams need automated surgical instrument tracking to prevent retained surgical items.

93. Women with prolonged second-stage labour need assisted fetal descent to prevent labor complications.
94. Laboratory technicians preparing microscopy slides need standardized slide preparation tools to improve diagnostic consistency.
95. Newborns exposed to meconium need early detection of meconium aspiration syndrome to enable immediate airway intervention.
96. Patients undergoing major surgery need early detection of blood-volume loss to prevent hypovolemic shock.
97. Patients receiving spinal anesthesia need precise anesthetic volume delivery monitoring to eliminate dosing errors.
98. Infants with congenital heart defects need portable cardiac monitoring to detect early deterioration.
99. Neonates receiving IV therapy need early detection of fluid extravasation to prevent tissue necrosis.
100. Patients arriving at emergency departments need rapid triage categorization to prioritize critically ill patients.
101. Premature infants post-discharge need remote physiological monitoring to enable early intervention for clinical decline.
102. Patients undergoing rehabilitation need objective functional progress measurement to guide physical therapy decisions.
103. Neonatal intensive care units need an affordable method to continuously monitor patient blood oxygen saturation to prevent hypoxia-induced organ damage.